UTSAV CHAUDHARY

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Senior **DevOps Engineer** with **7+ years** of experience building, automating, and operating **production-grade AWS and Azure infrastructure** in **regulated utility**, **energy**, and **enterprise consulting** environments, growing from build, release, and backend engineering into full platform ownership. Proficient in **Kubernetes and Helm**, **Infrastructure as Code** with **Terraform** and **Ansible**, **CI/CD** with **Jenkins**, **GitHub Actions**, and **Azure DevOps**, **GitOps** with **Argo CD**, and **multicloud deployments** across Dev, Test, UAT, QA, and Production, leading **disaster recovery**, **observability**, and **incident response** that meet **SLA**, **RTO**, and **RPO** targets.

Technologies

Cloud & Infrastructure: AWS (EC2, EKS, Lambda, RDS, S3, VPC, IAM, CloudWatch, API Gateway, Secrets Manager, Route 53, ALB/ELB, KMS), Azure (App Services, AKS, Azure DevOps, Azure SQL, Application Gateway, Front Door, Key Vault, Entra ID), Docker, Kubernetes, Helm, Rancher, Argo CD

IaC, Automation & Operating Systems: Terraform, Ansible, Puppet, Chef, CloudFormation, Bash, Python, PowerShell, YAML, Nginx, Cron, Linux (Ubuntu, CentOS, RHEL), Windows Server

CI/CD & Version Control: Jenkins, GitHub Actions, Bitbucket Pipelines, AWS CodePipeline, Azure DevOps, Git, GitHub, GitLab, Bitbucket, AWS CodeArtifact

Monitoring & Logging: Prometheus, Grafana, ELK Stack, CloudWatch, Application Insights, Datadog

Networking & Security: VPCs, VNets, Security Groups, NSGs, IAM Policies, DNS, VPN, LDAP, RBAC, SSL/TLS, OAuth 2.0, NERC CIP, ISO 27001, PCI, OWASP, vulnerability assessments, secrets management, key rotation

Practices & Development Tools: GitOps, canary releases, environment promotion, incident response, disaster recovery, Python, TypeScript, FastAPI, Node.js, React.js, Next.js, PostgreSQL, MongoDB, Redis, Kafka, Jira

Experience

Sr. DevOps Engineer, Hybrid, **MidAmerican Energy**, TX — Feb 2025 - Present

- Collaborated with platform and compliance teams on a **multicloud AWS/Azure** strategy, deploying **Kubernetes** workloads on **EKS/AKS** with **Helm** across **Dev** through **Production** for regulated **utility** platform services.
- Supported **GitOps** releases with **Argo CD** and **Rancher**, integrating **Azure Entra ID** for **RBAC** and **NERC CIP**-aligned audit logging.
- Maintained version-controlled **IaC** with **Terraform** and **Ansible**: **AWS VPCs**, **security groups**, **IAM policies**, **RDS**, **Azure VNets**, **NSGs**, **Key Vault**, **Azure SQL**.
- Built **observability** with **Prometheus**, **CloudWatch**, and **ELK** for **grid telemetry** and **smart meter (AMI)** pipelines; joined **on-call incident response** with **runbooks** restoring **outage management** to **SLA**.
- Built **CI/CD** in **GitHub Actions** and **Bitbucket Pipelines**: unit tests, **Docker** builds, **security scanning**, **Terraform** gates, **environment promotion**; led **design** and **code reviews**.
- Containerized **Python/Node.js** microservices and **ML load forecasting** on **EKS/AKS** with **HPA** scaling; automated **S3/Blob** and **billing database** backups per **data retention** rules.
- Managed secrets for **ERP**, **customer information system (CIS)**, and market data via **Secrets Manager/Key Vault** with **PCI**-aligned controls; led **root cause analysis** and **Python/Bash cost control** automation.

DevOps Engineer, On-Site, **Chevron**, GA — Jan 2024 - Jan 2025

- Supported production **AKS** clusters running **upstream and field operations workloads** behind **Application Gateway** with a distributed team, maintaining **99.9% availability** for **drilling reporting** services.
- Supported **monitoring** with **Application Insights**, **Azure Monitor**, and **Grafana**, contributing to **MTTD** dropping from **30** to **5 minutes** across **refinery maintenance** workflows.
- Assisted with migrating legacy apps to **cloud-native AKS**, improving **resource utilization 40%**, while **canary releases** via **Azure DevOps** traffic splitting cut deployment incidents **75%**.
- Maintained **Terraform** modules for **AKS**, **Azure SQL**, **Front Door**, **VNets**, **NSGs** with remote state and drift detection; automated **DR** to meet **RPO 10** and **RTO 30 minute** targets.
- Supported **environment promotion** across **Dev** through **Prod** with **Azure DevOps** approval gates and **Git workflows**, containerizing **FastAPI** and **Node.js telemetry APIs** with **automated testing**.
- Applied **Key Vault** encryption and key rotation for **proprietary seismic and operational data**; led **root cause analysis** and **vulnerability assessments** with **OWASP** findings.

DevOps Engineer CO-OP, On-Site, **Midh Technologies**, TX — June 2023 - Dec 2023

- Managed **GPU backed Kubernetes clusters** on **AWS EKS** for ML workloads with cost optimization through node pool configuration, resource quotas, and auto scaling policies.

- Designed high-performance inference infrastructure deploying **FastAPI** endpoints behind **AWS API Gateway** with **IAM authentication**, request throttling, and **Lambda** integrations for scalable ML inference.
- Cut manual server configuration time by **50%** with **Ansible playbooks**, and built production **CI/CD** in **GitHub Actions** and **Azure DevOps** with tests, **Docker** builds, **security scanning**, and **pytest** gates to **EKS**, **Amplify**, and **App Service**.
- Provisioned **AWS** and **Azure** environments with **Terraform** (**EKS**/**AKS**, **API Gateway**/**Azure API Management**, **S3**, **PostgreSQL**) enabling repeatable **environment promotion**, with **RBAC**, **IAM role**, and **VPC** security hardening across layers.
- Operated scheduled **Python ETL** jobs loading **PostgreSQL** and **MongoDB**, with containerized runs, retries, and **CloudWatch** alerting on failures.

DevOps/Platform Engineer, On-Site, **Accenture**, India June 2021 - July 2022

- Delivered platform engineering for **enterprise client accounts** in a global consulting engagement, managing hybrid **on premise Linux servers** (Ubuntu) with **Docker** deployments, hardened per **ISO 27001**-aligned standards with firewall rules, **SSH** keys, patching, and automated backups.
- Designed self hosted **CI/CD** with **GitHub Actions runners** and **Jenkins** on client infrastructure, building production grade pipelines for automated testing, **Docker** builds, and multi-environment deployments with standardized **Git branching strategies**, artifacts, and release tagging.
- Established **Nginx** reverse proxy infrastructure with **SSL/TLS** termination for client-facing applications at **250ms** average response time and high availability, keeping **WebSocket** streaming stable, and deployed **Prometheus**/**Grafana observability**.
- Containerized client **FastAPI** and **React** applications and scheduled **Python** scraping/**ETL** jobs (**Selenium**, **Pandas**) with **Docker** and **Cron**, automating **IT services** operations with **Python/Bash** and cutting manual overhead **60%**.

Software Engineer / DevOps, Remote, **VRoKCs**, MO March 2020 - May 2021

- Managed production infrastructure on **Azure**, deploying **Flask web applications** on **Azure App Services** with high availability, auto scaling, **Azure Monitor** alerting, and deployment slots for zero-downtime swaps.
- Developed **CI/CD** using **Azure DevOps** for **ML lifecycle** management, automating training and deployment workflows with **Azure Blob Storage** artifacts, reducing deployment cycles by **40%**.
- Designed containerized infrastructure with **Azure Container Instances** and **Docker** for **Flask** and **TensorFlow Lite** inference, keeping image upload, webcam streaming, and live prediction endpoints responsive with dev-prod parity.
- Added **automated testing** and **secure coding** checks to release pipelines for critical-path **Flask** APIs, partnering with ML engineers on **Python** preprocessing for data collection and datasets.

Software Developer / Build & Release, On-Site, **Wonderbiz.in**, India March 2019 - Feb 2020

- Set up **Jenkins** build jobs and **Git** versioning with release checklists for **environment promotion** across Dev, Test, and Production for **Industrial IoT** sensor ingestion, **asset tracking**, and dashboard services.
- Deployed and maintained **Python Flask** and **Node.js REST APIs** and the **Angular analytics dashboard** on **Linux** servers behind **Nginx**, with service restarts, log rotation, and health checks.
- Automated **SQL Server** backups, **sensor data** cleanup, and reporting with **Python** and **Bash** scripts scheduled through **Cron**, with **automated testing** and **secure coding** checks as release gates.

Web/Game Developer, On-Site, **underDogs Gaming Studios**, India Jan 2018 - Feb 2019

- Contributed to backend API infrastructure for gaming services, integrating and optimizing the **Quarters API** for secure transactions across multiple games, with **load balancing** and **API rate limiting** for high availability.
- Optimized real time client server communication in **Python** and **C#**, connecting external services (**payments**, **inventory**, **shipping**) to internal game systems with consistent API patterns.
- Applied **API security**, logging, and integration **automated testing**, and stood up release and hosting for game backends with **Git** source control and **health checks**.

Education

State University of New York at Binghamton Aug 2022 - Dec 2023
Master of Science in Computer Science 3.4/4.00 GPA

Certifications

Microsoft Certified: DevOps Engineer Expert View Credential
AZ-400: Designing and Implementing Microsoft DevOps Solutions
AZ-900: Microsoft Certified: Azure Fundamentals
Google IT Automation with Python (Specialization) View Credential